PalashSetu (पलाश सेतु) — SIH 2026 Master Pitch Deck & Script

Complete Slide-by-Slide Content & Word-for-Word Presenter Script (6-Slide Limit)

Problem Statement SIH 26042 • Department of School Education & Literacy, Govt. of Jharkhand • Team Psyduck

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⏱️ Team Pitch Strategy (5-Minute Hard Stop)

Slide #	Slide Title	Speaking Time	Main Concept to Nail
Slide 1	Title & Team Intro	**20 sec**	Who we are, problem statement, and core mission.
Slide 2	Problem, Solution & Uniqueness	**60 sec**	Comprehension shock, 5 tools, and modular tribal framework.
Slide 3	Technical Approach (2-Tier & Pipeline)	**80 sec**	2-Tier system, 4-tier NLP engine, and acoustic TTS compiler.
Slide 4	Feasibility, Viability & Mitigations	**50 sec**	₹0 recurring cost, runs on ₹5,000 tablets, multi-teacher auth.
Slide 5	Impact, Policy Alignment & Alphabet FLN	**60 sec**	NEP 2020 Clause 4.11, NIPUN Panchaadi & Ol Chiki alphabet.
Slide 6	Research, References & Live Demo	**30 sec**	AI4Bharat paper, NCERT MTB-MLE, and QR code to GitHub repo.

📌 SLIDE 1: TITLE & TEAM INTRODUCTION

📄 SLIDE CONTENT (What Goes on the Screen)

- **Project Title: PALASHSETU (पलाश सेतु)**
- **Subheading:** AI-Powered Offline MTB-MLE Assistant for Tribal Classrooms
- **Problem Statement:** SIH 26042 • Govt. of Jharkhand (School Education & Literacy Dept.)
- **Team Badge:** Team Psyduck
- **Team Members & Core Roles:**
  - Member 1: Team Lead & Mobile System Architect (Capacitor & Android)
  - Member 2: NLP & Linguistic Engine Lead (IndicTrans2 Distillation)
  - Member 3: Speech Audio & Acoustic Engineer (`santaliSpeech.ts`)

- Member 4: Curriculum & NIPUN Bharat Research Lead (JCERT Data)
  - Member 5: Frontend UI/UX Developer (React & Touch Controls)
  - Member 6: Quality Assurance & Test Verification Lead (44-Test Suite)
- **Logos:** Smart India Hackathon 2026, Ministry of Education, Govt. of Jharkhand

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🗣️ PRESENTER SCRIPT (Word-for-Word What to Say Out Loud)

> *"Respected judges, greetings of the day. We are **Team Psyduck**, and today we present **PalashSetu** — an AI-powered, 100% offline classroom companion built for Mother Tongue-Based Multilingual Education in tribal primary schools.*
>
> *In India, language should be a bridge to learning, not a barrier. Under Problem Statement 26042, we have engineered a standalone digital assistant that empowers Hindi-medium primary teachers to communicate, teach, and assess indigenous tribal children directly in their mother tongue — with zero internet, zero recurring cloud costs, and running entirely on low-cost government tablets."*

📌 SLIDE 2: PROBLEM STATEMENT, PROPOSED SOLUTION & UNIQUENESS

📄 SLIDE CONTENT (What Goes on the Screen)

Box 1: PROBLEM STATEMENT

- **MTB-MLE Language Divide:** Primary teachers instruct in standard Hindi, while tribal children speak indigenous mother tongues at home, causing early comprehension shock and dropouts.
- **Tribal Dialects Neglected:** Mainstream AI (Google, OpenAI) fails on tribal languages due to scarce digital corpora, non-standard scripts (Ol Chiki, Warang Chiti, Tolong Siki), and zero native OS voice support.
- **Total Connectivity Blackout:** Remote tribal schools & Anganwadis run at **0% internet** — cloud translation APIs are not an option.
- **Low-Cost Hardware Constraints:** Budget government tablets (2GB RAM, Android 9) cannot run heavy neural networks locally.
- **Flagship Target — Santali:** Selected for Phase 1 (~1.2M children in Jharkhand's Santhal Pargana & Kolhan), representing the highest linguistic isolation barrier.

Box 2: OUR PROPOSED SOLUTION (5 Core Pedagogical Tools)

- ➊ **Bidirectional Voice Bridge:** Real-time Teacher Hindi Speech-to-Voice + Interactive Student Ol Chiki Tap-to-Respond Interface.
- ➋ **NIPUN Bharat Lesson Studio:** 36 structured, 5-part Panchaadi lesson plans (Balvatika to Class 3) including dedicated Ol Chiki alphabet lessons.
- ➌ **Illustrated Tribal Flashcard Engine:** 30+ interactive decks (96+ cards) with 3D flip-and-reveal, visual aids, and native phonetics.
- ➍ **Dynamic Bilingual Worksheet Generator:** Infinite randomized drills with 1-tap printable A4 worksheets for offline paper distribution.

- ❺ **State Textbook Localizer (JCERT):** Dual-column side-by-side textbook reader with paragraph-level native audio triggers.

Box 3: WHY DIFFERENT / MODULAR INNOVATION

- **Modular Language Adapter:** Decouples UI from language data — easily add Ho, Mundari, and Kurukh by plugging in vocabulary arrays without core code changes.
- **Knowledge Distillation:** Compresses AI4Bharat's 320M-param IndicTrans2 neural model into an ultra-compact offline lexicon (7,500+ curated entries in Phase 1).
- **Acoustic-Phonetic TTS Bridge:** Solves the global absence of tribal TTS engines by mapping indigenous scripts to phonetic Indic phonemes for native offline vocalization.
- **4-Tier Resilient Fallback:** Phrase Match ➔ Regex Patterns ➔ Grammar Particles ➔ Phonetic Transliteration (proper names and novel words never fail).

Box 4: VALUE & TARGET SPECIFICATIONS

- **< 250 ms:** Target Voice Latency (comfortably beating the 3.0s SIH SLA requirement).
- **< 100 MB RAM:** Target Memory Footprint (engineered for budget 2GB RAM government tablets).
- **₹0 Recurring:** 100% free and open-source (no cloud hosting, no GPU servers, no API billing).
- **Scalability Roadmap:** Phase 1 (Live): Santali (Ol Chiki) • Phase 2 (Next): Ho & Mundari • Phase 3: Kurukh & Kharia.

🗣️ PRESENTER SCRIPT (Word-for-Word What to Say Out Loud)

> *"Judges, imagine a 6-year-old Santhali child on their first day of primary school in Dumka. At home, they speak Santali. But the moment they enter the classroom, the teacher speaks only Hindi, and the books are in Devanagari. This linguistic shock leads to cognitive fear, silence, and eventually, dropping out.*
>
> *Commercial solutions like Google Translate fail completely here: first, these rural schools have **zero internet**; second, Android has **no voice support for Santali**; and third, low-cost ₹5,000 government tablets cannot run heavy deep-learning models.*
>
> *To solve this, we built **PalashSetu**. It is not just a translator — it is a complete 5-in-1 classroom companion: featuring a real-time bidirectional voice bridge, 36 NIPUN Bharat structured lessons that even teach the Ol Chiki alphabet, interactive flashcards, infinite printable worksheets, and official JCERT state textbooks localized side-by-side.*
>
> **Most importantly, our architecture is **modular**. While we built and validated our Phase 1 pilot for **Santali (Ol Chiki)**, the framework is language-agnostic. We can scale to **Ho, Mundari, and Kurukh** simply by plugging in their vocabulary maps without modifying the app code."**

📌 SLIDE 3: TECHNICAL APPROACH & 2-TIER SYSTEM

📄 SLIDE CONTENT (What Goes on the Screen)

Part 1: THE 2-TIER SYSTEM ARCHITECTURE

- **Tier 1: Classroom Edge Node (Daily Tablet App):**

- 100% Standalone APK on ₹5,000 Android Tablets (React 18, TypeScript, Capacitor 6.1).
  - Operates in **100% Airplane Mode** (Zero internet, zero cloud servers, zero DB bloat).
  - Sub-millisecond in-memory translation via 7,503-word lexicon.

- **Tier 2: District Administrative Hub (Quarterly BRC Server):**
  - Central Python / FastAPI + SQLite server at the District Education Office.
  - Used for AI4Bharat IndicTrans2 model distillation, textbook authoring, and quarterly sync when tablets visit Block Resource Centres.

Part 2: CATEGORIZED TECH STACK

- **Client UI:** React 18.3, TypeScript 5.x, Vite 5 (SPA), High-Contrast Touch UI (48px+ targets).
- **Native Mobile:** Capacitor 6.1 Bridge, Android SDK (API 24–34), Hardware GPU Acceleration.
- **NLP & Translation Core:** AI4Bharat IndicTrans2 distillation, 7,503-Word In-Memory Lexicon, 4-Tier Rule Engine.
- **Speech Synthesis & Audio:** Custom `santaliSpeech.ts` Acoustic Compiler, Android Native `hi-IN` TTS, Web Audio SFX (`sfx.ts`).
- **Local Persistence:** Capacitor Preferences, SHA-256 local PIN encryption (`authService.ts`).

Part 3: END-TO-END DATA FLOW (Horizontal Pipeline)

```
[ Step 1: Input ]     ➔   [ Step 2: NLP Parser ]    ➔   [ Step 3: Audio Gen ]   ➔   [ Step 4: Output ]
Teacher speaks Hindi      4-Tier On-Device Engine:      Acoustic TTS: Ol Chiki       Instant spoken Santali
via tablet mic;           T1 Direct Match ➔ T2 Regex     mapped to Indic phonemes;     audio + dual-script
captured by offline       ➔ T3 Grammar ➔ T4 Script        dispatched to native          visual display on
SpeechRecognizer.         Transliteration fallback.     Android hi-IN TTS engine.    tablet screen.
```

🗣️ PRESENTER SCRIPT (Word-for-Word What to Say Out Loud)

> *"Now, let us examine our technical approach. How do we make state-of-the-art multilingual AI run offline on a 2GB RAM budget tablet?*
>
> *We designed a **Hybrid 2-Tier Architecture**:*
> *For daily teaching in remote jungle schools, the tablet operates as an **Autonomous Edge Node** — 100% offline, zero internet, zero cloud server. But for administrative curriculum updates, we maintain a **Central FastAPI Hub** at the District Headquarters for quarterly content synchronization.*
>
> *To eliminate heavy 2GB neural models from the tablet, we used **Knowledge Distillation**: we extracted 5,448 Santali vocabulary tokens from AI4Bharat's 320M IndicTrans2 model and compiled them into an ultra-fast, in-memory **7,500+ word lookup lexicon**.*
>
> *Our translation pipeline uses a **4-Tier Resilient Engine**: it checks direct matches first, applies regex classroom patterns, parses grammar postpositions like 'khon' (from) and 'habij' (until), and uses Tier 4 character transliteration for unseen student names like 'Rahul'. It never crashes and never returns a blank screen.*
>
> *For speech output, we solved a major global barrier: Android has zero native voice support for Santali. Our custom `santaliSpeech.ts` **compiler** maps Ol Chiki syllables into phonetic Indic sound equivalents in real time, articulating authentic Santali voice through Android's built-in offline engine at an ideal 0.85x classroom speed."*

📌 SLIDE 4: FEASIBILITY, VIABILITY & RISK MITIGATION

📄 SLIDE CONTENT (What Goes on the Screen)

Quadrant 1: TECHNICAL FEASIBILITY

- **Runs on Existing Hardware:** Engineered specifically for ₹5,000 government-issued tablets (2GB RAM, Android 7.0–9.0).
- **No Specialized Compute:** Requires no GPU, no NPU, and no cloud server; runs completely in memory.
- **Ultra-Low Resource Footprint:** Operates within **< 100 MB RAM** and **< 15 MB storage**, leaving 95% of tablet resources free.
- **Deterministic Reliability:** Zero LLM hallucinations; all output is linguistically and pedagogically verified.

Quadrant 2: OPERATIONAL & CLASSROOM FEASIBILITY

- **Zero Learning Curve:** 1-tap walkie-talkie mode designed for non-tech-savvy teachers in rural schools.
- **Shared Tablet Multi-Teacher Profiles:** Built-in **Local SHA-256 PIN Auth** (`authService.ts`) enables multiple teachers to share one tablet while preserving individual grade, school, and district settings.
- **A4 Printable Handout Exporter:** Generates ready-to-print paper worksheets and lesson plans for classrooms without individual student tablets.
- **Curriculum Compliance:** Pre-mapped to official **JCERT textbooks** and **NIPUN Bharat FLN learning outcomes**.

Quadrant 3: FINANCIAL & ECONOMIC VIABILITY

- **₹0 Recurring Cloud Cost:** No server subscriptions, no database hosting, no API token billing.
- **Zero Infrastructure Investment:** Deploys onto hardware already distributed under Samagra Shiksha Abhiyan / e-Vidyavahini.
- **Frictionless Offline Distribution:** Lightweight APK distributable via Bluetooth, SD cards, or USB drives in 2G areas without internet usage.

Quadrant 4: CHALLENGES & RISK MITIGATION TABLE

Potential Challenge	Impact Level	Our Engineering Mitigation
No Native Santali Voice in Android	🔴 Critical	Built custom `santaliSpeech.ts` **acoustic compiler** utilizing native Indic phoneme mapping.
No Offline Santali ASR for Students	🟡 Medium	Engineered **Interactive Tap-to-Respond Cards** (Balvatika/Class 1) with instant Hindi voice playback. (Santali voice ASR is on v2.0 roadmap).
Unseen Names & Novel Words	🟡 Medium	**Tier 4 Script Transliteration** converts proper nouns into Ol Chiki character-by-character.
Multi-Dialect Tribal Regions	🟢 Low	**Modular Language Adapter** allows adding Ho, Mundari, or Kurukh without touching core code.

🗣️ PRESENTER SCRIPT (Word-for-Word What to Say Out Loud)

> *"Turning to feasibility and viability: how do we know this will succeed on the ground?*
>
> *First, **Technically**: PalashSetu requires no expensive new hardware. It is compiled as a lightweight native Android APK that runs on existing 2GB RAM school tablets already distributed under government schemes like e-Vidyavahini.*
>
> *Second, **Operationally**: In rural schools, two or three teachers often share a single government tablet. We built a local, PIN-authenticated multi-profile service (`authService.ts`). Teacher Sunita in Class 1 and Teacher Ramesh in Balvatika can switch profiles on the same tablet with their own customized district and class settings. Furthermore, because children cannot stare at a screen all day, our app exports **1-tap printable A4 worksheets** for physical classroom distribution.*
>
> *Third, **On Real-World Challenges**: When evaluating student voice input, we recognized that Android has no offline Santali speech-recognition engine, and 5-year-old tribal children cannot reliably operate microphones. We solved this with an **Interactive Tap-to-Respond Interface**: children tap visual Ol Chiki response cards, and the tablet speaks Hindi to the teacher! Spoken Santali ASR is slated for our v2.0 roadmap.*
>
> *Finally, **Financially**: Commercial cloud AI translation APIs cost money per audio minute and per word translated. Across 1.2 million tribal children, cloud billing runs into crores. PalashSetu operates at **absolute zero recurring cost** to the state government — zero servers, zero API tokens, and zero data charges."*

📌 SLIDE 5: IMPACT, POLICY ALIGNMENT & ALPHABET FLN

📄 SLIDE CONTENT (What Goes on the Screen)

Left Zone: MEASURABLE CLASSROOM IMPACT

- **Eliminates Early-Grade Comprehension Shock:** Bridges the home mother tongue with the school medium, preventing fear and early dropouts.
- **Accelerates Foundational Literacy & Numeracy (FLN):** 36 structured Panchaadi lessons directly target Grade 3 reading and math fluency benchmarks.
- **Direct Ol Chiki Alphabet Teaching:** Dedicated curriculum units teaching **Ol Chiki Letters 1–10**, **Letters 11–20**, and **CVC Word Blending** (`ᱫ + ᱟ + ᱜ = ᱫᱟᱜ` Water, `ᱩ + ᱞ = ᱩᱞ` Mango).
- **Promotes Constitutional Script Pride:** Legitimizes the official **Ol Chiki script (`ᱚᱞ ᱪᱤᱠᱤ`)** in mainstream state classrooms, boosting student engagement and parental trust.

Right Zone (Top): TARGET OUTCOME METRICS (4 Metric Badges)

- 🎓 **1.2+ Million Children:** Primary school demographic benefited across Santhal Pargana and Kolhan divisions.
- 📉 **30–40% Reduction:** Projected decrease in early-grade dropouts attributed to language barrier alienation.
- ⚡ **< 250 ms Latency:** Real-time walkie-talkie communication enabling natural, fluid dialogue.
- 💰 **100% Budget Efficient:** Zero recurring server or network overhead for state education departments.

Right Zone (Bottom): NATIONAL POLICY & SCALABILITY ROADMAP

- **NEP 2020 Compliance (Clause 4.11):** Executes the mandate that primary education until Grade 5 be imparted in the mother tongue.

- **NIPUN Bharat Mission:** Follows the 5-step *Panchaadi* pedagogy: *Adhiti* ➔ *Bodh* ➔ *Abhyas* ➔ *Prayog* ➔ *Prasar*.
- **Scalability Roadmap:**
  - **Phase 1 (Live Pilot):** Santali (Ol Chiki Script) — 7,500+ lexicon, 36 lessons, 8 JCERT textbooks.
  - **Phase 2 (Immediate Target):** Ho (Warang Chiti) & Mundari (Jharkhand & Odisha border districts).
  - **Phase 3 (Statewide Expansion):** Kurukh/Oraon (Tolong Siki) & Kharia.

🗣️ PRESENTER SCRIPT (Word-for-Word What to Say Out Loud)

> *"The true measure of any educational innovation is its ground impact. What changes inside the classroom?*
>
> *First, it ends the traumatic **comprehension shock** that causes tribal children to drop out in Grades 1 and 2. When a child hears their mother tongue spoken warmly from their teacher's tablet, the classroom becomes an inviting space of trust.*
>
> *Second, it directly fulfills the national mandates of **NEP 2020 Clause 4.11** — which states that primary education must be in the child's home language — and the **NIPUN Bharat FLN mission**. Our 36 lessons directly follow the government's 5-step Panchaadi pedagogy: Connect, Explore, Explain, Practice, and Assess.*
>
> *Crucially, we don't just translate words — our Lesson Studio includes **dedicated Ol Chiki alphabet lessons**: teaching the first 10 letters, letters 11 to 20, and CVC word blending, so children learn to read and write their indigenous script systematically.*
>
> *Third, our impact extends far beyond Santali. While Santali represents our Phase 1 pilot covering 1.2 million children, our modular adapter architecture is designed to expand across Jharkhand's tribal belt — moving to **Ho in Kolhan**, **Mundari in Khunti**, and **Kurukh in Ranchi and Gumla**.*
>
> *PalashSetu preserves indigenous linguistic heritage while seamlessly scaffolding children into standard Hindi literacy."*

📌 SLIDE 6: RESEARCH, REFERENCES & DEMO

📄 SLIDE CONTENT (What Goes on the Screen)

Left Zone: RESEARCH FOUNDATIONS & SCIENTIFIC CITATIONS

- **1. AI4Bharat IndicTrans2 Consortium (Gala et al., 2023):**
  "IndicTrans2: Towards High-Quality and Accessible Machine Translation for all 22 Scheduled Indian Languages."
  (Provided baseline `sat_Olck` token vocabularies and validation pairs for distillation).
- **2. UNESCO & NCERT Guidelines on MTB-MLE (2021):**
  "Mother Tongue-Based Multilingual Education in Tribal Primary Schools: Policy and Pedagogical Frameworks."
  (Grounded our dual-column side-by-side reading layout and oral-first interaction model).
- **3. NIPUN Bharat Mission Framework (Ministry of Education, 2021):**
  "National Initiative for Proficiency in Reading with Understanding and Numeracy Guidelines."
  (Guided our 36 structured lessons, competency matrices, and numeracy drills).

Right Zone: OFFICIAL CURRICULUM, REPOSITORY & VERIFICATION

- **Official State Curriculum Data:**
  - **JCERT (Jharkhand Council of Educational Research & Training):** Official state primary textbooks localized into Ol Chiki.
  - **Unicode Consortium (ISO 15924 - `Olck`):** Standard character range `U+1C50–U+1C7F` used for deterministic transliteration.
- **Open-Source Repository & Deliverables:**
  - **GitHub Repository:** `github.com/puneetmehta288/PalashSetu`
    (Full source code, 7,500+ lexicon, test suites, and Android build scripts).
  - **Live Web Evaluation:** `palash-setu.vercel.app` (simulates on-device tablet execution).
  - **Standalone Android Package:** `PalashSetu-v1.0-debug.apk` (4.44 MB).
  - **Automated Test Suite:** 44/44 passing assertions via `test_offline_engine.js`.
- **[QR Code on slide pointing to GitHub / Live Demo]**

🗣️ PRESENTER SCRIPT (Word-for-Word What to Say Out Loud)

> *"To conclude, our technical methodology and curriculum design are anchored in peer-reviewed research and official government frameworks.*
>
> *Our linguistic engine builds upon the groundbreaking work of AI4Bharat's IndicTrans2 consortium from IIT Madras, while our pedagogical design adheres strictly to the NCERT MTB-MLE guidelines and official JCERT state textbooks.*
>
> *Our entire solution is fully built, tested, and open-source. The Android APK is only 4.4 MB, and our automated offline test suite passes 44 out of 44 linguistic assertions with sub-millisecond execution.*
>
> *Judges can scan the QR code on the screen right now to explore our GitHub repository and test the live engine.*
>
> *With PalashSetu, we are turning Jharkhand's state flower — the Palash — into a living digital bridge of multilingual education. Thank you, and we are now open for questions."*

🛡️ THE 5 CRUCIAL JUDGE QUESTIONS & DEFENSE SCRIPTS

Q1: "*How does this work offline without any internet?*"

> ***Answer:***
> *"Sir, we use Knowledge Distillation. Instead of running a heavy 2GB neural model on the tablet, we pre-distilled 5,448 Santali vocabulary tokens from IndicTrans2 into a 7,500-word in-memory hash table compiled directly into the app bundle. The dictionary, 36 lessons, and 8 textbooks are all packaged inside the 4.4 MB APK. Translation is an algorithmic in-memory lookup that takes less than a millisecond with zero network calls."*

Q2: "*Android doesn't have a Santali voice. How do you pronounce Ol Chiki?*"

> ***Answer:***
> *"That was our biggest engineering innovation. Passing raw Ol Chiki Unicode (`U+1C50-U+1C7F`) to Android TTS results in complete silence. We created a custom acoustic compiler (`santaliSpeech.ts`) that maps Ol Chiki syllables into phonetic Indic sound equivalents in real time. We then dispatch this phonetic string to Android's built-in offline `hi-IN` TTS voice at 0.85x speed, producing clear, authentic Santali speech completely offline."*

Q3: "*Does student voice recognition work in Santali?*"

> ***Answer:***
> *"Sir, Android currently has no offline Santali speech-recognition acoustic model. Furthermore, 5-year-old tribal children in noisy classrooms cannot reliably operate speech microphones. For Student-to-Teacher interaction, we engineered an **Interactive Tap-to-Respond Interface**: children tap visual Ol Chiki response tiles, and the tablet translates and speaks Hindi to the teacher. Direct Santali voice ASR via on-device quantized IndicWav2Vec is planned for our v2.0 roadmap."*

Q4: "*Why do you have a backend folder if the tablet is offline?*"

> ***Answer:***
> *"PalashSetu is a **Hybrid 2-Tier Architecture**. The tablet is the autonomous classroom edge device for daily teaching with zero server dependency. The FastAPI backend is the central administrative hub at the District Education Office used to distill new model tokens, author state curriculum, and push quarterly content updates when teachers visit the Block Resource Centre."*

Q5: "*Can this be used in schools where children speak Ho or Mundari?*"

> ***Answer:***
> *"Yes, absolutely! Our architecture is built as a **Modular Language Adapter** that decouples the UI from the linguistic data. Santali in Ol Chiki is our Phase 1 flagship pilot to prove the pipeline. To add Ho (Warang Chiti) or Mundari, we simply plug in their token lexicon and acoustic phoneme map. The user interface, lesson studio, worksheet generator, and audio synthesizer require zero code modifications."*